

Research Associate at the Max Planck Institute for Intelligent Systems in Tübingen, working on 3D digital humans in the team of Michael J. Black. My research centers on parametric body models (SMPL-X, MANO, and FLAME) and their use in human-object interaction, video diffusion models, and face reconstruction. Day-to-day, I work with PyTorch, building data and rendering pipelines, then fine-tuning and evaluating models.

EXPERIENCE

Max Planck Institute for Intelligent Systems, Perceiving Systems

02/2024 – Present

Research Associate

Tübingen, Germany

Projects

- First-authored InterDyn (CVPR 2025): fine-tuned a video diffusion model to generate controllable object dynamics from a driving signal, showing that large video models act as implicit neural physics engines
- Contributed to OFER (CVPR 2025): diffusion-based 3D face reconstruction from a single occluded image; captures the problem's multi-hypothesis nature by generating multiple plausible expressions rather than a single estimate
- Built an end-to-end training pipeline for egocentric human motion generation models using the EgoExo4D dataset. Implemented data storage/loading and distributed training/evaluation on H100 GPUs (unpublished)
- Optimized a multi-view geometry & optimization-based multi-view FLAME fitting pipeline (internal)
- Engineered an agentic visual navigation system pairing a frozen VLM with a motion-diffusion prior, to drive SMPL-X avatars through scanned 3D scenes (ScanNet) (internal)

Talks

- Spotlight talk at the Agents in Interaction: From Humans to Robots (H2R) workshop at CVPR 2025.

Loyens & Loeff

06/2023 – 08/2023

Working Student, Technology & Innovation

Amsterdam, The Netherlands

Supported the adoption of AI-powered legal-tech tools (including automated drafting) at a leading law firm. I also researched the legal-AI market to advise on solution selection for specific business cases.

Innovation Centre for Artificial Intelligence

08/2022 – 03/2023

Communication Manager (self-employed)

Amsterdam, The Netherlands

Grew the public profile of UvA's AI innovation center. I led interview campaigns with AI researchers, helped organise national AI events, and produced and distributed all web, LinkedIn, and social content.

Perron038

02/2021 – 05/2021

Machine Learning Internship

Zwolle, The Netherlands

Prototyped a non-destructive method to detect leakage invisible from the outside of sealed detergent-pod containers using active thermography. Co-designed a cobot-driven capture-and-heating rig for repeatable imaging, captured a thermal image dataset, and demonstrated reliable leak detection.

Tembo Group

05/2021 – 08/2021

Design Science Internship

Kampen, The Netherlands

Investigated the quality trade-offs of cross-silo horizontal federated learning for industrial machinery manufacturers via a systematic literature review. I designed and validated a framework for navigating competing requirements through a UTAUT-based case study.

LIDL Netherlands Headquarters

07/2018 – 03/2019

Customer Service Representative

Huizen, The Netherlands

Handled 100+ requests per day by phone and email, resolving service issues at LIDL's national headquarters.

PUBLICATIONS

- **InterDyn: Controllable Interactive Dynamics with Video Diffusion Models** [Paper](#) — [Website](#)
R. Akkerman, H. Feng, M. J. Black, D. Tzionas, V. Fernández Abrevaya
in Computer Vision and Pattern Recognition (**CVPR**), 2025
- **OFER: Occluded Face Expression Reconstruction** [Paper](#) — [Website](#)
P. Selvaraju, V. Fernández Abrevaya, T. Bolkart, R. Akkerman, T. Ding, F. Amjadi, I. Zharkov
in Computer Vision and Pattern Recognition (**CVPR**), 2025
- **[Re] Explaining in Style: Training a GAN to explain a classifier in StyleSpace** [Paper](#)
N. Van der Vleuten, T. Radusinović, R. Akkerman, M. Reksoprodjo (equal contribution)
in ReScience C, 8(2), 2022; ML Reproducibility Challenge (**MLRC**) 2021

EDUCATION

- University of Amsterdam, Master in Artificial Intelligence** 09/2021 – 08/2024
- Research-centered program — 3.8/4.0 GPA
 - Thesis: Synthesizing Object Dynamics from Human Motion with a Video Generative Prior (8/10) [↗](#)
- University of Twente, Bachelor in Technology and Liberal Arts & Sciences** 09/2018 – 08/2021
- Focused on machine learning and data science — 4.0/4.0 GPA
 - Honours Programme: Processes of Change (additional 30 ECs)
 - Thesis: Non-functional Requirements, Solutions and Interdependencies in Cross-Silo Horizontal Federated Learning

VOLUNTEERING

- CVPR & Eurographics** 2025 – 2026
- Peer Reviewer*

Reviewed conference submissions in computer graphics and computer vision.

- University of Twente Innovation Fellows** 10/2019 – 03/2024
- Fellow / Coordinating Team* *Enschede, The Netherlands*
- Directed a team of 12 students, overseeing strategic campus initiatives and co-managing an annual operating budget of €60,000; organised the 24-hour Twente-4-Hours event (KLM, Strategyzer, Stanford, FH Salzburg) and mentored new Fellows cohorts at TU Hamburg and Erasmus Rotterdam through Design Thinking training.

- DIH-HERO** 02/2021 – 08/2021
- Grants & Contracts Support Officer* *Enschede, The Netherlands*
- Supported coordination of a €16M Horizon 2020 action building a pan-European network of healthcare-robotics Digital Innovation Hubs across 17 partners, contributing to communications, events, and operations.

- Autumn Challenge** 08/2020 – 01/2021
- Coach*
- Advised an international student team developing a Care Hotel concept report.

- Stichting Huizer Botters** 01/2017 – 08/2018
- Secretary, De Bramzijgers* *Huizen, The Netherlands*
- Co-established “Bramzijgers”, the foundation’s youth wing, to pass on traditional botter craftsmanship and engage young members in preserving Huizen’s maritime heritage.

EXPERIENCE WITH

- **Stack:** Python; PyTorch, Hugging Face Transformers, Diffusers, PyTorch3D, PyRender/EGL, smplx, NumPy, OpenCV, project-aria; distributed multi-GPU training on A100/H100 via HTCCondor
- **Models:** Stable Video Diffusion, ControlNet, motion-diffusion priors (Kimodo, HunyuanMotion), vision-language models (Qwen-VL family), egocentric human motion generation (UniEgoMotion), Pixel3DMM
- **Datasets:** EgoExo4D, DexYCB, ScanNet, FaceScape, AMASS, PICO-db, DAMON, FaMoS, HM3D (Habitat), CLEVRER, Something-Something-v2, FaceExploration (MPI internal 3D face dataset)
- **3D & digital humans:** SMPL-family body models (SMPL-X, SMPL-H, MANO, FLAME), multi-view geometry & optimization-based multi-stage FLAME fitting, 2D/3D landmark detection
- **Languages:** English (proficient), Dutch (native), German (beginner), Spanish (learning)